# R Data types

# unit 1 (Chapter 2)

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# R operators

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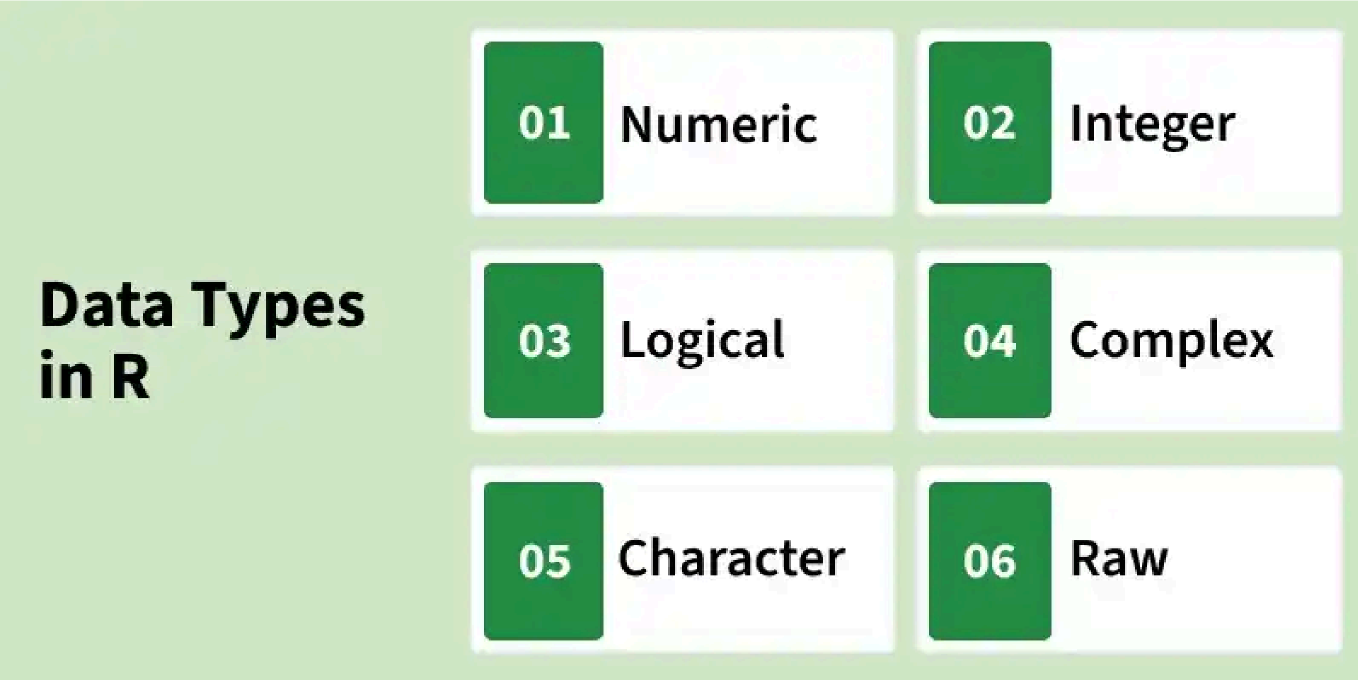
# R Data Types




- Logical: `TRUE / FALSE`
- Integer: `3L`
- Numeric (double): `10.5`
- Complex: `1+2i`
- Character: `"12-04-2020"`
- Raw

# Checking Data Types with `is.*`

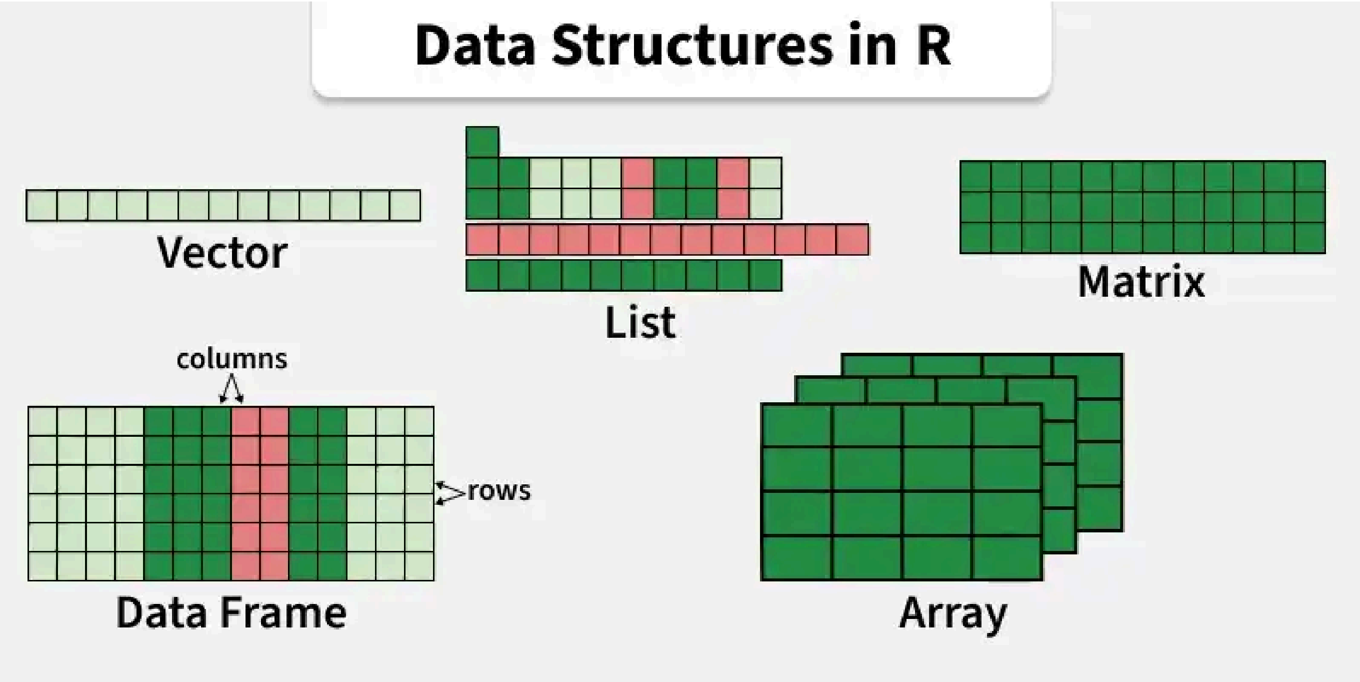
```
print(is.logical(TRUE))
print(is.integer(3L))
print(is.numeric(10.5))
print(is.complex(1+2i))
print(is.character("12-04-2020"))

print(is.integer("a"))
print(is.numeric(2+3i))
```

## Output:

```
[1] TRUE
[1] TRUE
[1] TRUE
[1] TRUE
[1] TRUE
[1] FALSE
[1] FALSE
```

# R Data Structures Overview




| Structure | Description |
|---|---|
| **Vector** | 1-D homogeneous data |
| **Matrix** | 2-D homogeneous |
| **Array** | n-D homogeneous |
| **List** | Heterogeneous, recursive |
| **Data Frame** | Tabular, heterogeneous |

All operations: **Create, Access, Insert, Update, Delete**.

# Vectors – Create & Access

## Create

```r
v <- c(10, 20, 30, 40)
v
```

```
[1] 10 20 30 40
```

## Access

```r
v[2]          # second element
v[c(1,3)]     # first and third
v[2:4]        # range
```

```
[1] 20
[1] 10 30
[1] 20 30 40
```

# Vectors – Insert, Update, Delete

## Insert

```
v <- c(v, 50)                 # at end
v <- append(v, 25, after = 2) # at position 3 (index after=2)
v
v<-append(v,1)
v
```

```
[1] 10 20 25 30 40 50
```

## Update

```
v[3] <- 100
v
```

```
[1]  10  20 100  30  40  50
```

## Delete (3rd element)

```
v <- v[-3]
v
```

```
[1] 10 20 30 40 50
```

## Length

```
length(v)
```

```
[1] 5
```

# conversion

| **Converts To** | **Example** |
|---|---|
| Numeric | `as.numeric("25")` |
| Integer | `as.integer(10.8)` |
| Character | `as.character(100)` |
| Logical | `as.logical(1)` |
| Factor | `as.factor(c("A","B","A"))` |
| Vector | `as.vector(matrix(1:4,2))` |
| Matrix | `as.matrix(df)` |
| Array | `as.array(matrix(1:4,2))` |
| List | `as.list(c(10,20,30))` |
| Data Frame | `as.data.frame(matrix(1:6,2))` |
| Date | `as.Date("2026-07-21")` |

# Matrices – Create & Access

## Create

```r
m <- matrix(1:9, nrow = 3)
m
```

```
     [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
```

## Access

```r
m[2,3]       # row 2, col 3
m[1,]        # row 1
m[,2]        # column 2
```

```
[1] 8
[1] 1 4 7
[1] 4 5 6
```

# Matrices – Insert, Update, Delete

## Insert Row / Column

```
#, eval=FALSE}
m <- matrix(1:9, nrow = 5)
m
```

```
     [,1] [,2]
[1,]    1    6
[2,]    2    7
[3,]    3    8
[4,]    4    9
[5,]    5    1
```

```
newrow <- c(10, 11, 12)
m <- rbind(m, newrow)
newcol <- c(13, 14, 15, 16)
m <- cbind(m, newcol)
m
```

```
              newcol
        1  6      13
        2  7      14
        3  8      15
        4  9      16
        5  1      13
newrow 10 11      14
```

## Update

```
#mat-update, eval=FALSE}
m[2,2] <- 100
m
```

```
                newcol
        1   6      13
        2 100      14
        3   8      15
        4   9      16
        5   1      13
newrow 10  11      14
```

## Delete Row (2nd) / Column (2nd)

```
#mat-delete, eval=FALSE}
print(m)
```

```
                newcol
        1   6      13
        2 100      14
        3   8      15
        4   9      16
        5   1      13
newrow 10  11      14
```

```r
m <- m[-2, ]   # delete row
print(m)
```

```
                 newcol
         1   6      13
         3   8      15
         4   9      16
         5   1      13
newrow 10 11      14
```

```r
m <- m[, -2]   # delete column
print(m)
```

```
             newcol
         1      13
         3      15
         4      16
         5      13
newrow 10      14
```

# Dimensions

```r
#mat-dim, eval=FALSE}
dim(m)
```

```
[1] 5 2
```

```r
nrow(m)
```

```
[1] 5
```

```r
ncol(m)
```

```
[1] 2
```

# Arrays – Create & Access

## Create (3-D)

```
arr <- array(1:24, dim = c(2, 3, 4))
arr
```

```
, , 1

     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

, , 2

     [,1] [,2] [,3]
[1,]    7    9   11
[2,]    8   10   12

, , 3

     [,1] [,2] [,3]
[1,]   13   15   17
[2,]   14   16   18

, , 4

     [,1] [,2] [,3]
[1,]   19   21   23
[2,]   20   22   24
```

## Access

```
#arr-access, eval=FALSE}
arr[1, 2, 3]      # single element
```

```
[1] 15
```

```
arr[, , 2]        # entire 2nd layer (matrix)
```

```
     [,1] [,2] [,3]
[1,]    7    9   11
[2,]    8   10   12
```

## Update

```
#arr-update, eval=FALSE}
arr[1, 1, 1] <- 100
arr[, , 2] <- matrix(0, 2, 3)   # replace layer
arr
```

```
, , 1

     [,1] [,2] [,3]
[1,]  100    3    5
[2,]    2    4    6
```

```
, , 2

     [,1] [,2] [,3]
[1,]    0    0    0
[2,]    0    0    0

, , 3

     [,1] [,2] [,3]
[1,]   13   15   17
[2,]   14   16   18

, , 4

     [,1] [,2] [,3]
[1,]   19   21   23
[2,]   20   22   24
```

## Dimensions

```
#, eval=FALSE}
dim(arr)
```

```
[1] 2 3 4
```

# Lists – Create & Access

## Create

```
student <- list(
  Name = "Ravi",
  Age = 21,
  Marks = c(80, 90, 85)
)
student
```

```
$Name
[1] "Ravi"

$Age
[1] 21

$Marks
[1] 80 90 85
```

## Access

```
student$Name
student[[2]]
student[["Marks"]]
```

```
[1] "Ravi"
[1] 21
[1] 80 90 85
```

# Lists – Insert, Update, Delete

## Insert

```
student$Branch <- "CSE"
student
```

## Update

```
student <- list(
  Name = "Ravi",
  Age = 21,
  Marks = c(80, 90, 85)
)
# list-update, eval=FALSE}
student$Age <- 22
student
```

```
$Name
[1] "Ravi"

$Age
[1] 22

$Marks
[1] 80 90 85
```

## Delete

```
#list-delete, eval=FALSE}
student$Marks <- NULL
student
```

```
$Name
[1] "Ravi"

$Age
[1] 22
```

## Names & Length

```
names(student)
length(student)
```

```
[1] "Name"   "Age"    "Branch"
[1] 3
```

# Data Frames – Create & Access

## Create

```r
#, eval=FALSE}
student <- data.frame(
  RollNo = c(101, 102, 103),
  Name   = c("Ram", "Ravi", "John"),
  Marks  = c(85, 90, 88)
)
student
```

```
  RollNo Name Marks
1    101  Ram    85
2    102 Ravi    90
3    103 John    88
```


```
  RollNo Name Marks
1    101  Ram    85
2    102 Ravi    90
3    103 John    88
```


## Access Column / Row

```r
student$Marks
```

```
[1] 85 90 88
```

```r
student[2, ]      # row 2
```

```
  RollNo Name Marks
2    102 Ravi    90
```

```r
student[, 2]      # column 2
```

```
[1] "Ram"  "Ravi" "John"
```

```r
student[1:2, ]    # first two rows
```

```
  RollNo Name Marks
1    101  Ram    85
2    102 Ravi    90
```

# Data Frames – Insert, Update, Delete

## Insert Column

```
#df-insert-col, eval=FALSE}
student$Grade <- c("A", "A+", "A")
student
```

```
  RollNo Name Marks Grade
1    101  Ram    85     A
2    102 Ravi    90    A+
3    103 John    88     A
```

## Insert Row

```
#df-insert-row, eval=FALSE}
student
```

```
  RollNo Name Marks Grade
1    101  Ram    85     A
2    102 Ravi    90    A+
3    103 John    88     A
```

```
class(student)
```

```
[1] "data.frame"
```

```
newrow <- data.frame(RollNo=104,Name = "Anil", Marks=45, Grade = "A+")
student <- rbind(student, newrow)
student
```

```
  RollNo Name Marks Grade
1    101  Ram    85     A
2    102 Ravi    90    A+
3    103 John    88     A
4    104 Anil    45    A+
```

## Update (change Marks of second row)

```
#df-update, eval=FALSE}
student$Marks[2] <- 95
student
```

```
  RollNo Name Marks Grade
1    101  Ram    85     A
2    102 Ravi    95    A+
3    103 John    88     A
4    104 Anil    45    A+
```

## Delete Column / Row

```r
#df-delete, eval=FALSE}
student$Grade <- NULL            # remove Grade
student <- student[-3, ]         # remove 3rd row
student
```

```
  RollNo Name Marks
1    101  Ram    85
2    102 Ravi    95
4    104 Anil    45
```

# Data Frames – Structure & Dimensions

## Structure

```
#df-str, eval=FALSE}
str(student)
```

```
'data.frame':   3 obs. of  3 variables:
 $ RollNo: num  101 102 104
 $ Name  : chr  "Ram" "Ravi" "Anil"
 $ Marks : num  85 95 45
```

## Number of Rows & Columns

```
#df-dim, eval=FALSE}
nrow(student)
```

```
[1] 3
```

```
ncol(student)
```

```
[1] 3
```

```
dim(student)
```

```
[1] 3 3
```

# Summary Table

| Structure | Create | Access | Insert | Update | Delete |
|---|---|---|---|---|---|
| **Vector** | `c()` | `v[i]` | `append()`, `c()` | `v[i] <- val` | `v[-i]` |
| **Matrix** | `matrix()` | `m[i,j]` | `rbind()`, `cbind()` | `m[i,j] <- val` | `m[-i,]`, `m[,-j]` |
| **Array** | `array()` | `arr[i,j,k]` | assign values | `arr[i,j,k] <- val` | (reassign) |
| **List** | `list()` | `$`, `[[ ]]` | `list$new <- val` | `list$name <- val` | `list$name <- NULL` |
| **Data Frame** | `data.frame()` | `$`, `[ ]` | `rbind()`, `cbind()` | `df[row,col] <- val` | `df[-row,]`, `df$col <- NULL` |